GLOBAL FINITE ELEMENT MODEL VERIFICATION AND VALIDATION PROCEDURE

OBJETO / *PURPOSE:*

The object of this document is to establish the basis for the verification, validation, storage and exchange of Aircraft major component Global FE models (GFEM) within Airbus DS Military Aircraft, and will be applicable to all current and future analysis structures and projects. The document identifies the Verification and Validation checks that must be performed on the FEM to assure a minimum quality level of the FE Analysis performed for structural analysis and structural dynamics.

The document includes the Verification forms that must be fulfilled for each GFEM that will be used for Stress and Fatigue Analysis

This Procedure completes the FEA Manual described in DPI-ADC-006

ALCANCE / *SCOPE:*

This DPI is applicable to all the Finite Element Analysis performed by

- # AIRFRAME

This Manual is applicable to all numerical simulations (linear and not linear) using finite element technique. The specific checks contained in the manual are oriented to MSC Nastran tool but the philosophy could be applied to any solver.

Aprobación Propietario / *Owner Approval:*

Nombre / *Name:* Oslé Dorremochea, Eduardo

Función / *Function*: #STRESS, METHODS & OPTMIZATION

Autorización / *Authorization:*

Nombre / *Name:* FAUSTE, José Luis

Función / *Function*: #Structural Analysis

Fecha / *Date:* *dd/MM/yyyy*

Vº Bº / Endorsement

Date: ... Name: Dr. Weishaeupl, Caroline Function: #Airframe	Date: ... Name: MENNLE, Edgar Function: #Structural Integrity
Date: ... Name: Daoud, Fernass Function: #Stress Methods and Optimization	Date: ... Name: Calero, Miguel Ángel Function: #Stress
Date: ... Name: Dr. Gómez-Escalonilla, Javier Function: #Fatigue & DTA	Date: ... Name: Dr. Climent, Héctor Function: #Structural Dynamics & Aerolasticity
Date: ... Name: ... Function: ...	Date: ... Name: ... Function: ...
Date: ... Name: ... Function: ...	Date: ... Name: ... Function: ...
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1 INTRODUCTION

1.1 PURPOSE

The object of this document is to establish the basis for the verification, validation, storage and exchange of Aircraft major component Global FE models within Airbus DS Military Aircraft, and will be applicable to all future analysis structures and projects.

Global FEM are the Finite Element Models devoted to perform stress analysis for Aircraft major structural components, let's say, the numerical models used to translate Aircraft Loads into structural internal loads.

The intention is to identify the Verification and Validation checks that must be performed on the FEM to assure a minimum quality level and reduce the common source of errors using these numerical models.

The correct application of this procedure will not guarantee that the Global FE model results are acceptable to the project, only that the results will reliably reflect the response of the structure and will not be affected by analysis anomalies.

This Procedure is complementary with the DPI-ADC-006.

1.2 SCOPE

This DPI is applicable to all the Global Finite Element Analysis performed by

- # AIRFRAME

The DPI is applicable to all GFEM developed in MSC.Nastran used internally in TAES, and also the GFEM provided to other organizations as TAEG. This Manual is applicable to all numerical simulations using linear finite element technique.

The specific checks contained in the manual are oriented to MSC.Nastran tool but the philosophy could be applied to any other FE solver.

For Detailed FEM applications (Linear and Non-linear) there are complementary guidelines that will be used according to the kind of analysis to be performed. See NT-T-ADP-10003-B for additional information.

This Procedure can be substitute by other similar requirements (as Airbus FEA Manual M2036) if required by project specifications.

2 FINITE ELEMENT ANALYSIS IN STRUCTURAL ANALYSIS

The main objectives of the Global Finite Element Model are to represent accurately the A/C stiffness and reproduce the load paths across the structure.

The results of the GFEM analysis are a critical input for Stress and Fatigue Certification Analysis. Due to that we must try to assure a high level of confidence on GFEM and reduce to a minimum the source of errors.

GFEM is a computational model, based on a specific FEA code (and one specific version of this code) used to represent an aircraft structure behavior.

According to the definition agreed in ASME V&V 10-2006, GFEM Verification is the process devoted to minimize and/or characterize the errors or limitations of the computational and mathematical model: modeling errors, code errors, results consistency, etc.

In the same reference, GFEM Validation is the process where the physics of the model is compared with experimental results, in order to assure that the physics are accurately represented by the GFEM.

2.1 Finite Element Analysis Code

The F.E. Analysis Code used for GFEM Analysis shall be MSC.Nastran.

The specific version of MSC.Nastran to be used will be determined and agreed by all parties for each major project, or major revision of project. The MSC.Nastran Version shall be specified in the Verification Forms.

2.2 Finite Element Model Definition

The component models supplied must assembled into a Complete Aircraft model without further work being necessary, and must have sufficient and well balanced detail to permit combined aeroelastic and strength optimization and, when assembled as part of the overall aircraft model, must provide the correct load paths, stiffness behavior and deflection of the complete model.

2.2.1 Components

The FEM of the complete aircraft shall comprise the following components:

- A complete full diameter fuselage
- Both Left and Right wings
- Both Left and Right Horizontal Tailplanes
- Vertical Tailplane
- Engine Pylons or Engine Mounting System (EMS)
- All movable surfaces (including actuators and attachments structures)
- Any other relevant element that could impact on A/C stiffness or load path as:

- External Radar Dome and corresponding struts
- Any relevant underwing pod and pylons
- Air Refueling Boom System
- Etc.

The following components will be supplied if available or required:

- Landing Gears
- All Fairings required for Dynamic Analysis

Each structural component shall be supplied as a separate MSC.Nastran bulk data file, additional data required to enable integration, verification and validation of the component should also be supplied as a further computer file.

The model supplied must contain the following data:

- The FE model data shall be supplied in N and mm.
- The FE model must include all the requisite attachments for the application of masses for Dynamic Analysis.
- The FE model must include the LIM to GFEM connection, in order to allow loads application.
- Sufficient comments on idealization (Boundaries, Spec., Features …).

2.2.2 Plane of Symmetry

In order to allow the generation of a half-aircraft model from the full aircraft model, no elements must cross the aircraft plane-of-symmetry.

All the DOF at the plane of symmetry must be independent.

For elements on or in the vertical plane-of-symmetry, they will be represented with complete properties, and it will be required to be divided by 2 when generating the half-model.

Doubled elements on the plane-of-symmetry may be used providing all parties involved are in agreement; but property references must be unique to that location to eliminate the problem of multiple addressing of property cards.

2.2.3 FEM Reference and Numbering Criteria

The NASTRAN "basic co-ordinate system" of the model shall be the basic aircraft co-ordinate system as defined in the DMU or DBD. An alternative master co-ordinate system may be used if agreed by all the departments/partners. This alternative reference must be clearly documented.

The model bulk data will be numbered according to a convention defined at the beginning of the project. Some example is shown in following tables, according to Light & Medium Transport A/C or Airbus derivatives (including A400M):

	Light & Medium A/C	Airbus deriv.
GRID	C YY ZZZ	C YYY ZZZ (*)
CORDxR/FORCE/MOMENT		CXXX
SPC/LOAD		CXXX
Element/Property/Material	CW YY ZZZ (*)	C X YYY ZZZ (*)
PCOMP/MAT8		C XX XXX (*)(**)

Table 2.1. Node and Element Identification ranges.

Where C represents the component identification number:

No element number should be duplicated, and the numbers assigned should be meaningful. Where possible the numbering logic should be retained between different versions of model for the same project.

The verification of the FE Identification Range must be checked and signed in Verification Form 3.

Notes:

(*) The range of numbering for the element, property, material and GRID identifications for the Fuselage will be changed to starting values of 1 (always) after the availability of a new MSC.Nastran verified version with extended PCOMP and MAT8 numbering (8 digits) and not further aeroelastic limitation.

(**) From Nastran 2001 the numbering restrictions on PCOMP cards will no longer exist.

3 GFEM VERIFICATION PROCESS

This chapter will establish the GFEM verification procedure. The chapter will describe also the forms that must be fulfilled in order to obtain GFEM verification.

A more detailed description of the process and possibilities will be described in chapter 7.

This chapter 3 contains the forms and also includes an example of Verification Forms fulfillment.

Every Form shall be signed by the engineer that has been performed the checks and shall be approved by Stress Methods and Optimization Team or authorized engineer designed by Airframe Organization.

Every GFEM used to support any stress and fatigue certification document must be verified according to this procedure and the forms must be available in an official document.

3.1 GFEM IDENTIFICATION AND VERSION

Every GFEM should have a unique identification and a version number. This GFEM identification must be used in the GFEM Description documents and for Stress and Fatigue analysis that are based on one specific GFEM.

The GFEM identification is a code of 16 alphanumeric characters (capital letters and numbers, including a '-') with the following structure:

1	2	3	-	4

Each specific field is described as follows

Field 1 corresponds to Project ID, as :

Project	Code	Project	Code
EF2000	EFA	A400M	A400M
A320	A320	C295	C295
Falcon 7X	F7X	A330/340	A330
Talarion	TAL	MALE 2020	MALE

Table 3.1 Project Code for GFEM Identification

For Future Projects this code must be fixed by Structural Analysis and Structural Integrity responsible.

Field 2: Identification of requested department: D ➔ Dynamic and Aeroelasticity

E ➔ Stress (Default, if GFEM is used for different disciplines)

F ➔ Fatigue

M ➔ Stress Methods

Field 3: Free characters.

Field 4: GFEM version (2 digits).

This identification will be inserted in all the Verification Forms and it will be part of the Main input file ID.

Example:

C295ETMBR2F23-01: C295 GFEM, for Stress Analysis, Military version, corresponding to Brazil 02 configuration, with Flap 23 deg, version 01.

3.2 GFEM VERIFICATION: REFERENCES (Form 1)

GFEM Verification Form 1 shall contain the general information with the references used to generate the GFEM:

- General description and Purpose of the GFEM. Components modeled, Units, MSC.Nastran version, etc.
- Basic Co-ordinate reference of the GFEM.
- Geometry CAD reference (CATPart, Product,…) used to generate the GFEM.
- Meshing Rules or reference
- Element Properties and Materials properties: source of properties and material data (DAMA, MMPDS, referenced document…)

3.3 GFEM VERIFICATION: Files Record (Form 2)

GFEM Verification Form 2 shall contain the list of all the MSC.Nastran input files that takes part of the GFEM and all the files used to complete the GFEM Verification process.

Every file included in the main input file deck shall be included in the list with a short description.

3.4 GEOMETRICAL CHECK LIST (Form 3)

A series of basic model data checks shall be performed on the analysis model. These checks shall include the following:

a) FEM Identification Criteria: Confirm that the criteria for FEM numbering, concerning Nodes, Elements, Properties, Materials, etc. are fulfilled.

b) Identification of Coincident Nodes and Elements. Free edge/face to confirm there are no voids in the model.

c) Verify 2D element normals to confirm consistency of 2D element orientations and G1-G2 direction.

d) Bending planes and/or offsets for element consistency.

e) Panel shape assessments for Aspect Ratio, Taper, Twist and Skew.

f) Identification and orientation of all co-ordinate systems.

g) Dependent/Independent node conflicts for Rigid Elements.

h) Degrees of freedom of the Model, to have an idea about the size of the GFEM

i) Computer, Operating System and MSC.Nastran version used for the analysis

j) MSC.Nastran Parameters used in the analysis

k) Maximum Diagonal Factor and singularity table, extracted form Nastran f06 output file.

l) Any other relevant item that must be highlighted: DMAPS, Nastran statement, etc.

3.5 STATIC ANALYSIS CHECK LIST (Form 4)

The Boundary conditions used for the Static Checks shall be described. A referenced document with plots and additional explanation can be used as description.

The applied load case used for static analysis checks must be a representative load case of the structure: it must be a limit or ultimate load case, representing some critical conditions. The load case shall be identified with NASTRAN SUBCASE and the load case identification.

The Form 4 can be included as many times as required for verifying statically the GFEM. Each form will be used for a representative load case. As examples:

- Fuselage: Internal Pressure, Max Bending, Max Torque.
- HTP/VTP/Wing: Max bending, max. torque, Maximum Hinge Moment at movable surface

OLOAD RESULTANT will be compared with SPCFORCE RESULTANT (with opposite sign). Difference in percentage shall be identified.

The grid points where the displacements and rotations in each direction are maxima absolute shall be identified.

The grid points where the SPCFORCES and Moments in each direction are maxima absolute shall be identified.

The Epsilon and external Work value shall be extracted for the evaluated Load Case from f06 file.

Any estrange behavior or remark shall be included in the Comments area.

3.6 DYNAMIC ANALYSIS CHECK LIST

Two different dynamic checks shall be performed to verify the model:

- Dynamic checks using corresponding density values in the Material cards of the NASTRAN deck.
- Dynamic using realistic mass distributions.

Remarks to be taken into account to perform dynamic checks:

a) These dynamic checks shall be made using SOL 103 of MSC.Nastran using the LANCZOS method.

b) Use the corresponding PARAM WTMASS value to assure that the unit of the frequencies obtained is Hz.

c) All analyses must be made with AUTOSPC = NO or "PARAM, AUTOSPC, NO". This is normal practice.

d) Translation singularities must be eliminated with the correct use of RBE elements. Any rotational singularities not constrained by using PARAM, K6ROT, 1. and PARAM, SNORM, 20. must be eliminated with SPC's or RBE's, as normal practice.

e) Under NO circumstances must users use field 8 of the GRID and GRIDSET cards to include permanent single point constraints. This is normal practice as the use of that field can cause the inadvertent grounding of masses during dynamic analyses or loads during static analyses if the node is used as a mass or loading point. Singularities should be eliminated via SPC1 or SPC cards.

f) The deflected shape of the model should be assessed, with drawings of the deformed structure, using the basic co-ordinate system. Areas where the displacement co-ordinate system is not the basic one should be specifically identified and plotted where possible.

3.6.1 Dynamic Checks using Density Properties on MAT Cards (Form 5)

These dynamic checks included in Form must be performed in order to get GFEM Verification. These dynamic checks with densities values try to identify mechanism or areas with low stiffness.

CHECK	Test Masses	Method	Frequencies	Boundary Conditions	Target
1	Densities in MATx cards	Lanczos	[-1, 1] Hz	free/free with actuators ($K_{act} \simeq 10^8$ N/m)	6 and only 6 rigid body modes (RBM) ($f_{RBM} < 0.01$ Hz) **Rigid Body Check (GROUNDCHECK)**
2	Densities in MATx cards	Lanczos	[0, 1] Hz	free/free with actuators disconnected	6 rigid body modes ($f_{RBM} < 0.01$ Hz) plus 0 Hz rotations (one per control surface)
3	Densities in MATx cards	Lanczos	[1, ∞] Hz 15 first modes	free/free with actuators	Identify Flexible Modes. Detect local or unrealistic modes detection
4	Densities in MATx cards	Lanczos	[0, 25] Hz	free/free with interfaces disconnected	Component isolated 6 RBM ($f_{RBM} < 0.01$ Hz) Identification of spurious links at interfaces

Table 3.2. Normal Modes Checks with Densities (Form 5)

- Check 1 shall include additionally GROUNDCHECK verification in order to check strain energy levels for G, N, F and A sets (see 7.5).
- Check 2: shall be performed if the GFEM contains movable surfaces. GFEM in free-free conditions with Actuators disconnected (movable surfaces: Ailerons, Elevator, rudder,

etc.). Normal Modes between [0, 1] Hz. The analysis shall obtain 6 and only 6 rigid body modes, with very low frequency (f_{RBM} < 0.01 Hz) plus one rigid body rotation per control surface.

- Check 4: shall be performed if the GFEM will be delivered to TAEGD (Dynamics and Aeroelasticity) for loads and aeroelastic calculations. GFEM in free-free conditions with disconnected interfaces (for the main components: Fuselage, Wing, HTP, VTP, control surfaces). Normal Modes between [0, 25] Hz. The analysis shall obtain 6 rigid body modes per component, with very low frequency (f_{RBM} < 0.01 Hz), and first flexible modes for isolated component.

Form 5 shall be fulfilled indicating the mass reference obtained in NASTRAN run, comparing it with Reference mass of the structure. The table shall contain the main frequency modes (in Hz) in each run.

Parameters, comments or remarks shall be included accordingly.

3.6.2 Dynamic Checks using CONM2 (Form 6)

These dynamic checks included in Form must be performed in order to get GFEM Verification. These dynamic checks with CONM2 values try to identify the real dynamic behavior of the GFEM and detect local mechanism or areas with low stiffness.

CHECK	Test Masses	Method	Frequencies	Boundary Conditions	Target
5	CONM2	Lanczos	[-1, 1] Hz	free/free with actuators ($K_{act} \simeq 10^8$ N/m)	6 and only 6 rigid body modes (RBM) (f_{RBM} < 0.01 Hz)
6	CONM2	Lanczos	[0, 1] Hz	free/free with actuators disconnected	6 rigid body modes (f_{RBM} < 0.01 Hz) plus 0 Hz rotations (one per control surface)
7	CONM2	Lanczos	[1, ∞] Hz 15 first modes	free/free with actuators	Identify Flexible Modes. Detect local or unrealistic modes detection
8	CONM2	Lanczos	[0, 25] Hz	free/free with interfaces disconnected	Component isolated 6 RBM (f_{RBM} < 0.01 Hz) Identification of spurious links at interfaces

Table 3.3. Normal Modes Checks with Densities (Form 6)

- Check 6: shall be performed if the GFEM contains movable surfaces. GFEM in free-free conditions with Actuators disconnected (movable surfaces: Ailerons, Elevator, rudder, etc.). Normal Modes between 0, 1] Hz. The analysis shall obtain 6 and only 6 rigid body modes, with very low frequency (f_{RBM} < 0.01 Hz) plus one rigid body rotation per control surface.

- Check 8: shall be performed if the GFEM will be delivered to TAEGD (Dynamics and Aeroelasticity) for loads and aeroelastic calculations. GFEM in free-free conditions with disconnected interfaces (for the main components: Fuselage, Wing, HTP, VTP, control surfaces). Normal Modes between [0, 25] Hz. The analysis shall obtain 6 rigid body modes per component, with very low frequency (f_{RBM} < 0.01 Hz), and first flexible modes for isolated component.

Form 6 shall be fulfilled indicating the mass reference obtained with the CONM2 cards in NASTRAN run, comparing it with Reference mass of the A/C or component. The table shall contain the main frequency modes (in Hz) in each run.

Parameters, comments or remarks shall be included accordingly.

3.7 THERMO-ELASTIC ANALYSIS CHECK LIST (Form 7)

To carry out this check, it is necessary to modify the thermal expansion coefficients by applying identical and uniform CTE to all MAT cards of the GFEM. The analyst must input a constant value for all of the $\alpha_{i,i}$ expansion coefficients and ensure that the $\alpha_{i,j}$ (where i≠j) coupling coefficients are set to zero.

Within the Nastran input deck, the reference temperature is set to a constant value for all materials. The parameter TEMP (INIT) = 20 and TEMP (LOAD) = 2 are inserted in the Case Control Section and similarly TEMPD, 20, 20 and TEMPD, 2, 120 into the Bulk Data Section. (The indicated values are set to perform a uniform ΔT = 100°C).

Upon completion of the modifications, the whole model will be run using MSC.Nastran SOL 101.

Setting in the Case Control parameter ESE=ALL allows the user to view the element stresses and identify those which fail to satisfy the Zero Stress criteria (values exceeding 1MPa).

Form 7 shall be fulfilled with the most relevant results of the analysis.

If the GFEM does not fulfill the thermoelastic check, tis must be stated in the comments.

3.8 QUALITY ASSURANCE FORMS

The quality Assurance Forms are shown in the next slides. The forms are included as attachments to this procedure.

An explanation about how to fulfill these forms and an example fulfilled is also included in section 7.7.

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION REFERENCES	PROJECT:
PART COMPONENT:	A/C COMPONENT:	PAGE: 1
GFEM IDENTIFICATION:		VERSION:

Finite Element Model general description:

Reference Coordinate System (FEM Component related to DBD/CATIA)

Geometry

Meshing

Properties

Materials

	Name (Dept.)	Signature	Date
Compiled by:			
Approved by:			

DPI-ADC-007 Issue F – Form 1

	GFEM VERIFICATION **FILES RECORD**	**PROJECT:**
PART COMPONENT:	**A/C COMPONENT:**	**PAGE:** 2
GFEM IDENTIFICATION:		**VERSION:**

COMPUTER DATASET / FILE NAME	DATE OF CREATION	COMMENTS/DESCRIPTION

	Name (Dept.)	Signature	Date
Compiled by:			
Approved by:			

DPI-ADC-007 Issue F – Form 2

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION **MODEL CHECK LIST**	**PROJECT:**

PART COMPONENT:	**A/C COMPONENT:**	**PAGE:** 3

GFEM IDENTIFICATION:	**VERSION:**

MODEL SIZE					
GRIDs	**1-D**	**2-D**	**3-D**	**0-D**	**MPC / RBE**

CHECK LIST	**DEPARTMENT RESPONSIBLE & PRINCIPAL DATA VALUE (If it is relevant)**	**SIGNATURE**
FEM Identification Criteria		
Coincident Nodes		
Coincident Elements		
Free Edges/Free Faces		
Element Warping		
Element Twist		
Interior Angles		
Distortion of Elements		
Connectivity		
Bending Planes (BAR&BEAMs)		
Offsets (all elements)		
Positive normal (QUAD4/TRIA3)		
Element Co-ordinate System		
Material Co-ordinate System		
Co-ordinate Systems		
Rigid Elements, MPCs		
Degrees of Freedom (KLL – F04)		

ANALYSIS DATA		
Computer	**Operating System**	**MSC.Nastran Version**

Max. Diagonal Factor		**Is Grid Point Singularity. Table Empty?**			
If YES		**If NO**	**Attached as page No.:**		

MSC.NASTRAN Parameters

COMMENTS:

	Name (Dept.)	**Signature**	**Date**
Compiled by:			
Approved by:			

DPI-ADC-007 Issue F – Form 3

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION **STATIC ANALYSIS CHECK LIST**	PROJECT:

PART COMPONENT:	A/C COMPONENT:	PAGE: 4

GFEM IDENTIFICATION:	VERSION:

BOUNDARY CONDITIONS DESCRIPTION:
(text, figure or references)

LOAD CASE DESCRIPTION:

Load Balance Checks	FX	FY	FZ	MX	MY	MZ
Applied						
Reaction (SPCs)						
Difference (%)						
Max. Displacements	TX	TY	TZ	RX	RY	RZ
GRID						
Value						
Max. SPCFORCES	FX	FY	FZ	MX	MY	MZ
GRID						
Value						
Epsilon			External Work			

COMMENTS:

	Name (Dept.)	Signature	Date
Compiled by:			
Approved by:			

DPI-ADC-007 Issue F – Form 4

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION ANALYSIS CHECK LIST: NORMAL MODES 1	PROJECT:
PART COMPONENT:	A/C COMPONENT:	PAGE: 5
GFEM IDENTIFICATION:		VERSION:

	G-Set	N-Set	F-Set	A-Set
Rigid Body Check				
	Check 1	**Check 2**	**Check 3**	**Check 4**
Mass Reference				
Mass Output from GPWG				
Mode Plots & Frequency List				

	Frequency [Hz]			
	Check 1	Check 2	Check 3	Check 4
Mode No:	(Densities in MATx) [-1,1] free/free with actuators	(Densities in MATx) [0,1] free/free without actuators	(Densities in MATx) [1,∞] free/free with actuators	(Densities in MATx) [0,∞] free/free with interfaces disconnected
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				

Comments:

	Name (Dept.)	Signature	Date
Compiled by:			
Approved by:			
Dynamic Acceptance:			

DPI-ADC-007 Issue F – Form 5

GFEM VERIFICATION ANALYSIS CHECK LIST: NORMAL MODES 2		PROJECT:
PART COMPONENT:	A/C COMPONENT:	PAGE: 6
GFEM IDENTIFICATION:	VERSION:	
BOUNDARY CONDITIONS DESCRIPTION: (text, figure or references)		

	Check 5	Check 6	Check 7	Check 8
Mass Reference				
Mass Output from GPWG				
Mode Plots & Frequency List				

	Frequency [Hz]			
	Check 5 (M1)	Check 6 (M2)	Check 7 (M2)	Check 8 (M2)
Mode No:	(Masses in CONMx) [-1,1] free/free with actuators	(Masses in CONMx) [0,∞] free/free without actuators	(Masses in CONMx) [1,∞] free/free with actuators	(Masses in CONMx) [0,∞] free/free with interfaces disconnected
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				

Comments:

	Name (Dept.)	Signature	Date
Compiled by:			
Approved by:			
Dynamic Acceptance:			

DPI-ADC-007 Issue F – Form 6

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION **THERMO-ELASTIC ANALYSIS CHECK LIST**		PROJECT:
PART COMPONENT:	A/C COMPONENT:		PAGE: 7
GFEM IDENTIFICATION:		VERSION:	

BOUNDARY CONDITIONS DESCRIPTION:
(text, figure or references)

Isostatic constraints

LOAD CASE DESCRIPTION:

Defined Temperature :

Load Balance Checks	FX	FY	FZ	MX	MY	MZ
Applied						
Reaction (SPCs)						
Difference (%)						
Stress	MAX	MIN				
ELEMENT						
Value						
Epsilon			External Work			

MSC.NASTRAN Parameters

COMMENTS:

	Name (Dept.)	Signature	Date
Compiled by:			
Approved by:			

DPI-ADC-007 Issue F – Form 7

4 GFEM VALIDATION PROCESS

GFEM Validation is performed when we can assure that the results of the Computational model (GFEM) reproduce accurately the results of some physical tests that are representative of the behavior of the A/C Structure.

Usually the validation is performed at two levels: Static and Dynamic

4.1 GFEM Static Validation

The static Validation is performed by correlating the GFEM Results with the results of Structural Tests, as Full Scale Tests (FST) or component Tests.

4.1.1 GFEM Boundary Condition

GFEM Boundary condition must be updated to be similar to the constraints used in the Static Tests.

Dummy elements can be introduced in the FEM representation, if needed: Movable surfaces, Landing gears…

4.1.2 GFEM Loading process

The loading process used in the Finite Element Model must be representative of the physics used in the Test.

The Load shall be introduced according with the directions and magnitudes used in Test. The load shall be linked to the structure trying to reproduce the load repartition mechanisms used in test.

4.1.3 Material Properties

The GFEM Material properties must be updated to the conditions of the test. In most of the cases the MAT card values (E, G, μ) are modified to get RT ambient values, at 20^{o}C.

4.1.4 Results to be Correlated

- Strain Values coming from strain gauges
- Displacements obtained by displacement captors
- Reactions
- Forces calibrated at different locations: Interfaces

For GFEM validation purposes we will select the strain gauges located at a ‘clean’ location, let’s say, locations where the GFEM are locally representative of the real structure.

Please avoid strain gauges located in panels with ply drop-off or thickness variation, or near cut-outs or secondary structure.

4.1.5 Correlation process

The validation is performed using the Correlation Plots. For Strain outputs, some figures shall be produced representing Strain(FEM) vs Strain (test) and plot according next figure.

Reference document Airbus.

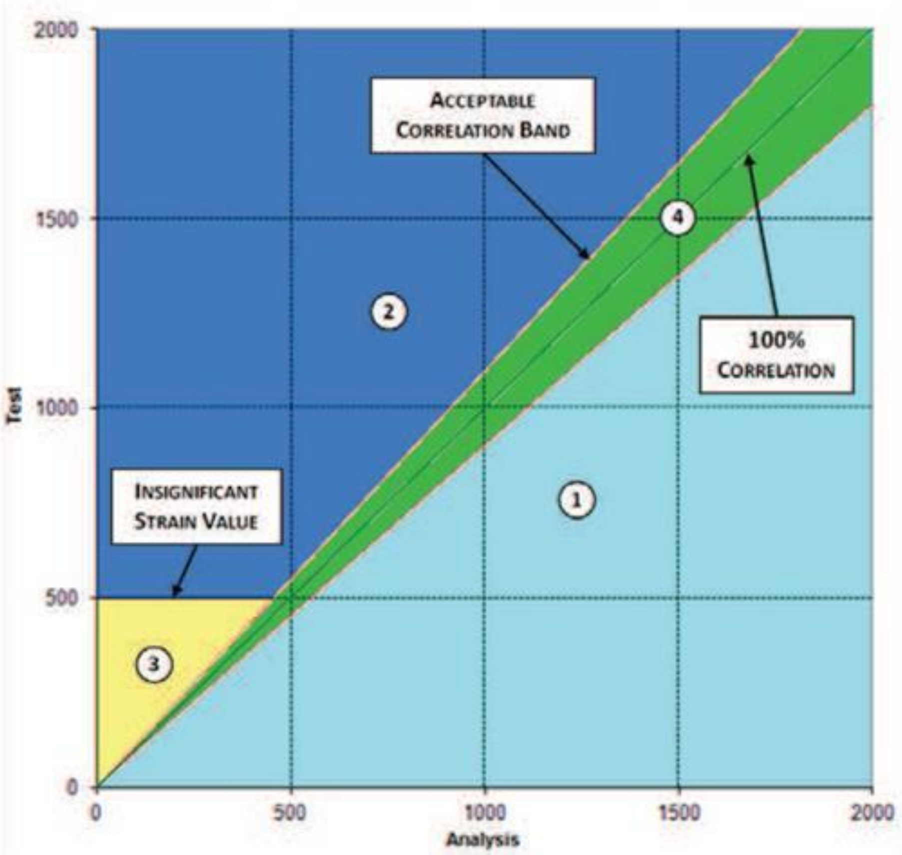


Figure 5.1. Strain Correlation Plots

Zone	Conclusion
1	The simulation (GFEM analysis) seems to be conservative compared to the test result and the analysis is validated by the test result accordingly. Adequate justification shall be established and the extent of the implications delimited.
2	The simulation prediction is not conservative compared to the test result. GFEM results shall be adjusted in order to cover the lack of conservatism or adequate justification shall be established and the extent of the implications delimited.
3	The prediction is not conservative compared to the test result. As it is a secondary or non-significant deflection/strain level, neither the analysis nor the ACD shall be adjusted.
4	As the delta between simulation and the test result is within an acceptable range of accuracy (typically ±10%), the analysis is validated.

Table 5.1. Conclusions from Correlation Results

Similar plots shall be performed with displacements captured in Test and equivalent obtained at GRID points in GFEM. For

After correlation, if GFEM needs some adjustment in order to match FST results, a new version of the GFEM will be performed.

The updated validated version of the GFEM shall be the official version for the upcoming analysis.

4.2 GFEM Dynamic Validation

GFEM is validated dynamically by means of Ground Vibration Test (GVT). This job is performed by Structural Dynamics and Aeroelasticity team, with support from Airframe teams.

During GVT, the flexible mode frequencies and mode shapes are captured from the real A/C. These are compared with the ones obtained by the dynamic model with the corresponding adjusted mass status. The correlation process is described in the flowchart included in Table 5.2.

To perform this validation, the GFEM must be completed with the corresponding Mass Model that reproduces the GVT configuration.

The Dynamic Updating Model is performed in different ways:

- Adding a Delta-Stick model: This is a suite of CBAR/CBEAMs elements added at the component elastic axis, that increases the stiffness to match GVT results. The stick model represents the additional stiffness provided by secondary elements that are not represented in GFEM: leading edge, trailing edge, fairings, etc

- Updating stiffness at interfaces between components: GFEM usually represents the critical load paths to get the interface loads in a conservative way. The fairings or secondary elements are not represented, but can modify slightly the modes frequencies. In order to match the frequencies the Dynamic GFEM can be modified at interfaces if it is required and justified.

Once validated, The Dynamic GFEM will be used as the reference Model for Dynamic Loads calculation and Aeroelastic Analysis.

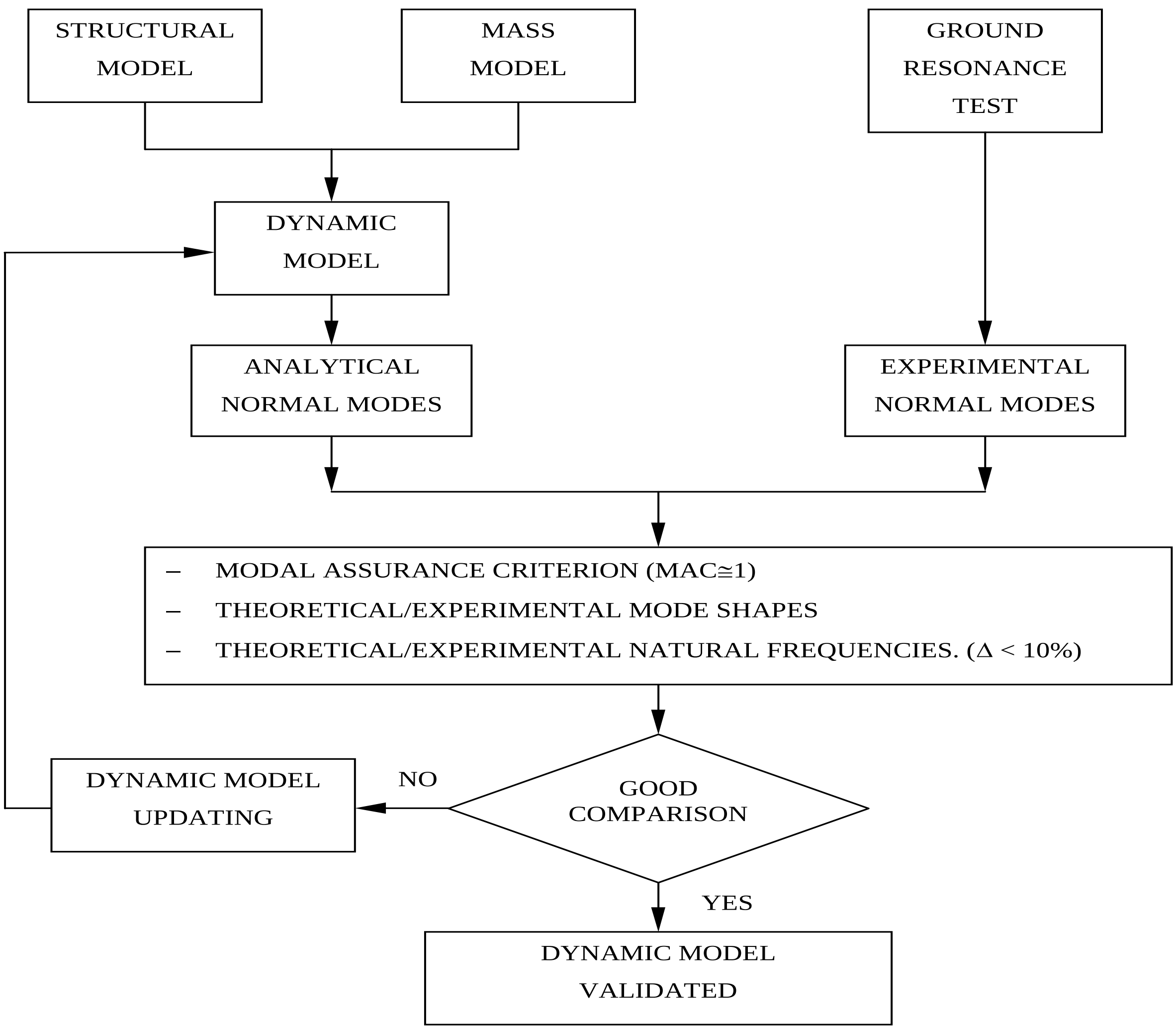


Figure 5.2. GFEM Dynamic Validation Process description

5 GFEM DOCUMENTATION

According to DPI-ADC-009, the Global FEM shall be documented. The document shall contain the description of the GFEM, Grids and Elements identification, properties and Materials used. All the main interfaces shall be described in detail.

The document shall have a specific chapter dedicated to Quality Assurance Forms. All the Forms fulfilled and signed shall be included in this chapter.

All the files containing the FEM data and FE Analysis output data shall be stored according with the procedure DP-A00-008.

The GFEM Validation shall be documented in a different document, including all relevant correlation plots and the proposed GFEM modifications, if exist.

6 GFEM UPDATING PROCESS

A GFEM shall be updated if:

- The Full Scale Test or GVT shows some areas with bad correlation between Simulation and test.
- Any modelisation error or improvement has been highlighted that have relevant impact on stiffness or load paths.

In this case a new version of the GFEM shall be created, and the version of the GFEM ID (section 3.1) will be increased in 1 digit.

The updated GFEM shall be verified according to this procedure and the Verification Forms shall be fulfilled, and stored with the GFEM updated. The new version will substitute the previous one, which will be archived.

All the departments affected shall be informed about the updated version.

A new Stress or Fatigue analysis, and a new GFEM document issue shall be generated if the amount of changes are relevant. This must be agreed by all the involved organizations.

7 GUIDELINES AND GENERAL EXPLANATIONS

7.1 Objective

The objective of this chapter is to clarify the correct application of the recommendations written in current procedure, and identify areas where it will be necessary to expect definitions of local variations. This chapter will contains some guidelines to correctly fulfill the GFEM verification Forms.

Current procedure outlines basic principles of good practice that should be applied to all forms of analysis including Linear Elastic. Some of the checks are required in order to use thr GFEM for Loads and Aeroelastic Analysis.

The same checks are required to be performed for the assembly of component models (i.e. Aircraft) as have been applied to the individual components models (i.e. wing or fuselage).

This chapter tries to provide a best practice method for performing the geometrical and static checks. These are only recommendations, the analysts may use any method that is considered suitable to achieve the required results and satisfactorily complete the checks.

7.2 NASTRAN Parameters

There are two ways of change or tune the NASTRAN element formulation:

- NASTRAN Statement: at the beginning of the NASTRAN Deck
- NASTRAN Parameters: Inserted in Case Control or Bulk data

The PARAM options quoted show best advice at the time of writing and in general should be adhered to. As analysis techniques change, the desired PARAM options may also change, and therefore it is important that the analysis PARAM configurations should be agreed at the launch meeting to ensure analysis compatibility.

With the command "ECHO = SORT(PARAM)" in the case control the PARAMs used are written in F06 file. It is recommended in order to check the PARAMs used and write them in corresponding Forms.

7.3 Guidelines for Geometry Checks (Form 3)

In order to fully verify model elemental geometric relationships it is "necessary" to use both a Pre-processing tool (Patran or Hypermesh) and MSC.Nastran in two different phases.

These two phases are not independent, but complement each other, and it is recommended that the checks using the pre-processing tool (Patran or Hypermesh) are performed in a first step, followed by checks using MSC.Nastran. When all the checks in this section have been performed, Form 3 (for geometry checks) must be completed. Before any specific data checks are performed, the MSC.Nastran bulk data deck should be read into pre-processing tool in order to identify any errors in element connectivity, numbering and entity duplication.

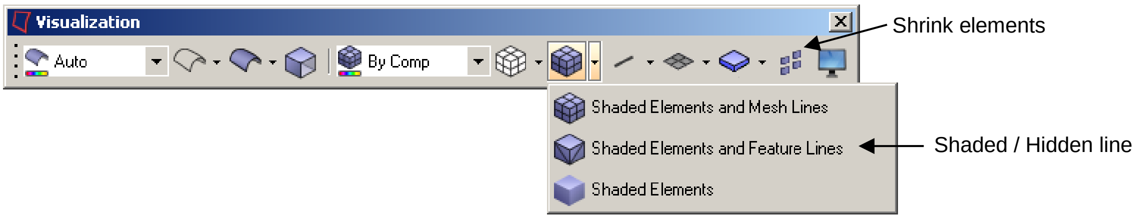
At this stage plots for the complete structure can be prepared for later use in the documentation. These plots also form part of the data checks, as they will illustrate fundamental misalignments between groups of nodes and elements.

7.3.1 Geometrical Checks using Pre-processing Tool

These checks should be performed using the "Finite Element" or "Utilities" menus within **Patran** set the "graphic utilities" to either "shrink", "hide line" or "shaded" mode and verify element connectivity and type.

These options are also available in Hypermesh within the visualization toolbar, as shown in the following figure.

Hypermesh. Visualization toolbar



The "Utilities Display" menu can be used to check Element Type, Property data, Material data and Element co-ordinate Systems by using "Show Element Info", whereas in Hypermesh, this information is available in the following menus:

- **Element Type:** Main Menu: 2D – Element Type – review
- **Property data:** Main Menu: Element color mode – By Prop Or locating it on the model tree on the Tab Area
- **Material data:** Main Menu: Element color mode – By Mat Or locating it on the model tree on the Tab Area
- **Element coordinate systems:** Menu Bar: Mesh – Assign – Element Material Orientation - review

In addition, all this information can also be obtained by directly checking the Nastran card with the “Card Edit” tool.

The following are the geometrical checks required:

7.3.1.1 Coincident Nodes

To be made in <u>Patran</u> using the command "Equivalence". This command removes, for one list, group, or the whole model, the nodes located in the same position, with a user defined tolerance, erasing the undesired free edges. The coincident nodes linked by ELAS, BUSH, MPC, and rigid elements are not eliminated by Patran. The erased nodes are highlighted and the command may be reversed with "UNDO".

To display the coincident nodes linked by ELAS and/or GAP elements, you may use the menu "Utilities/Display/Mark 1D Elements". By putting the elements with length smaller than a user defined tolerance value in a list, the user may create a list of nodes related to these elements using the menu "List" command.

The same functionality is available in Hypermesh under Main Menu: Tool – Edges – Select Elems / comps – Equivalence. The tool 'Display Element Handles' from the 'Display' Toolbar can be used to check for the presence of 1D elements, like ELAS, GAP or BUSH.

7.3.1.2 Free Edges

Patran:

The command "Verify/Element/Boundaries" is used to check the free edges of surfaces and identify disconnects between 2D elements. This command shows the free edges or free faces of the elements included in the group or groups shown in the screen, which are not connected to other elements of the model. The analyst must confirm that any free edges identified are valid. A plot should be obtained illustrating the free edge boundary, and it is recommended that this is drawn using 2D (QUAD4 and TRIA3) elements only as 1D elements will always show as free edges.

Hypermesh:

"Main Menu: Tool – Edges – Select elems / comps – find edges". The edges are saved as 1D elements in a separate component named '^edges'

7.3.1.3 Coincident Elements

Patran:

The command "Verify/Element/Duplicates" in Patran is used to check for coincident elements. This command shows and lists all the elements with the same nodal connectivity. The command will erase those duplicate elements with the highest identification or vice-versa.

Hypermesh:

In Hypermesh, this functionality is available under: "Main Menu: Tool – check elems – duplicates" the elements can be erased on the erase menu. The analyst must confirm that only planned duplicate elements exist.

7.3.1.4 Normals Verification

Patran:

Command "Verify/Element/Normals" is used to verify element normals. The option "draw normal vector" displays the direction of element normals for 2D elements included in the current group. The command allows the user to reverse the normal of some elements to match the normal of the correctly aligned elements.

The QUAD4 element co-ordinate system used by Patran to obtain the element normal is shown in Figure 3.

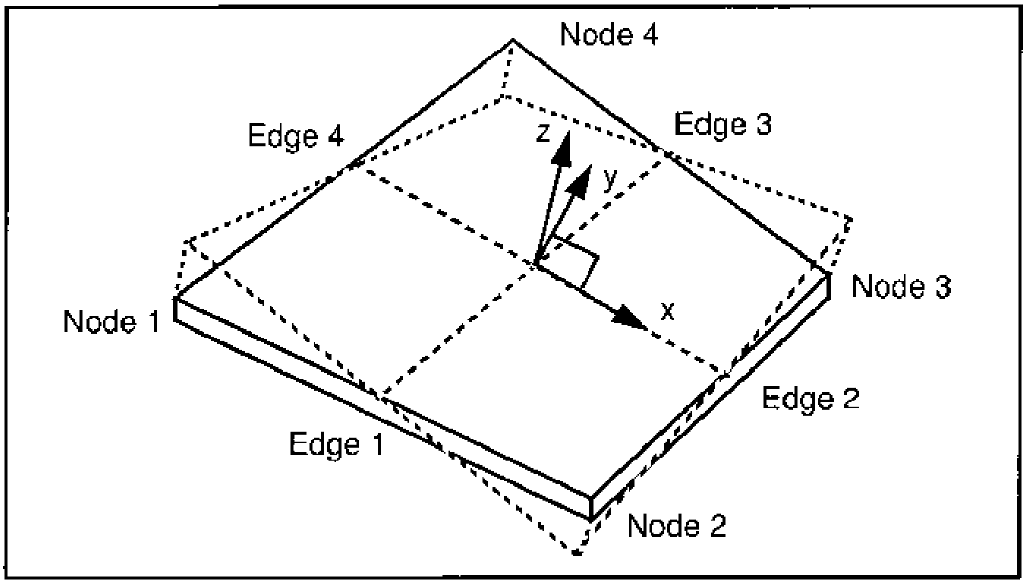


Figure 7.1. QUAD4 element co-ordinate system in Patran

Hypermesh:

“Main Menu: Tool – normals – select elems / comps – vector display”. The menu gives the possibility to adjust the normal for single or groups of elements.

7.3.1.5 Element Distortions

The command "Finite Element/Verify" in Patran or “Tool – check elems – 2-d” in Hypermesh is used to check for element distortions. This command measures the deviation between the actual element and the ideal one. The worst element distortions must be located outside the zones with large stress concentration or high stress gradient.

Two parameters are checked separately:

7.3.1.5.1 Aspect Ratio (TRIA3 & QUAD4).

Aspect ratio is defined as longest side / shortest side for both CTRIA and CQUAD elements. Very high aspect ratio should be avoided, although it is no longer true that accuracy degrades rapidly with aspect ratios as it once did with some of the obsolete elements.

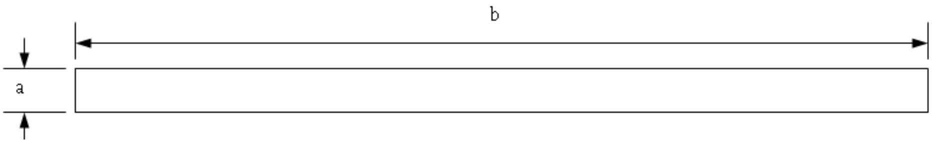


Fig. 7.2. Element with high aspect ratio

7.3.1.5.2 Taper (QUAD4)

Nastran finds the taper of quadrilateral elements by treating each node as the corner of a triangle (using one of the quad's diagonals as the triangle's third edge). The areas of each of these four "virtual" triangles are compared to one half of the total area of the quadrilateral element to produce a ratio; the largest of these ratios is then compared to the tolerance value. A value of 1.0 is a perfect quadrilateral, and higher numbers denote greater taper.

Taper recommended values are between 0.8 and 1.

Hypermesh reports an equivalent taper: this means that the smallest area ratio found is subtracted from 1, so that 0 represents a perfect quadrilateral element instead of 1.0, and greater deviation from 0 indicates greater taper. Recommended values on Hypermesh are between 0 and 0.2.

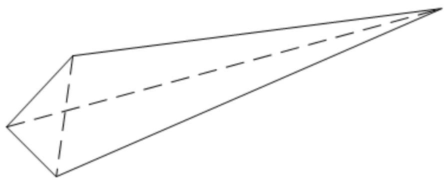


Fig. 7.3. Element CQUAD4 with high taper

7.3.1.6 Element Warping

In Patran, the command "Finite Element/Verify/QUAD/Warp" is used to check the element warping, this functionaly in Hypermesh is also available in the "Tool – check elems – 2-d" menu. This factor measures the amount from which an element deviates from being planar.

Warping angles smaller than 10 degrees are recommended.

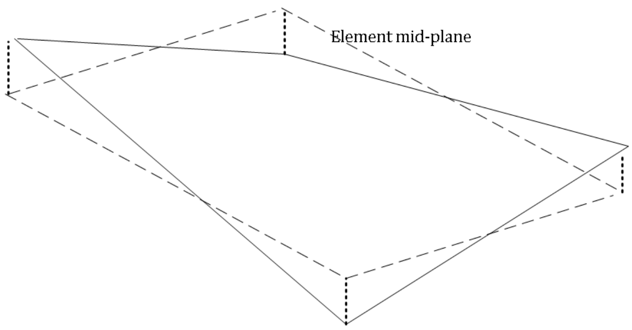


Figure 7.4. Highly warped element

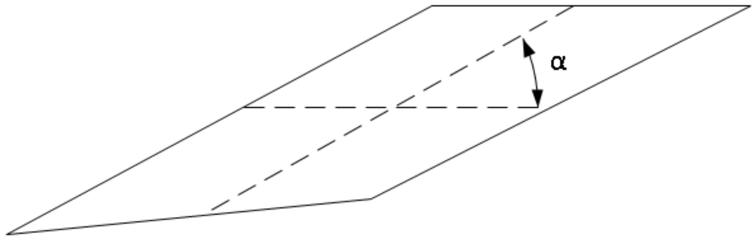
7.3.1.7 Internal Angles (Skew) and Skew Factor

The command "Verify/QUAD or TRIA/Skew" is used to check element internal angles in Patran; in Hypermesh is also available in the "Tool – check elems – 2-d" menu. This command presents the largest element internal angle.

Skew in trias is calculated by finding the minimum angle between the vector from each node to the opposing mid-side and the vector between the two adjacent mid-sides at each node of the element. Ninety degrees minus the minimum angle found is reported as the skew.

Skew in quads is calculated by finding the minimum angle between two lines joining opposite mid-sides of the element. Ninety degrees minus the minimum angle found is reported.

Hypermesh reports the angle directly while Patran reports the ratio between this angle and 90º. This ratio must be in the range between 0 and 0.33. Elements with internal angles higher than recommended values should not be located inside the zones with high stress concentration or a high stress gradient.



Fig, 7.5. Highly skewed element

7.3.1.8 Spring (CELAS / CBUSH) Elements

It must be confirmed that all CELAS or CBUSH elements have zero length and have coincident nodes. It must be checked that the nodes linked by CELASx elements have the same Analysis Coordinate systems. This operation may be combined with the operations described in the section Coincident Nodes (see A.5.1.1.1). In Patran, the menu "Utilities/Display/Mark 1D Elements" is used to put elements with null length, i.e. smaller than a user defined tolerance value, into a group. Using the menu "Utilities/Display/Show/Element info", you may check the stiffness (K), the co-ordinates of the nodes, the co-ordinate system for displacements of those nodes, and the degrees of freedom affected for all elements within the group.

7.3.1.9 Rigid Elements and MPCs

MPCs, and Rigid Elements, are equations that define relationships between degrees of freedom of nodes. It must be confirmed that the relationship between the degrees of freedom used are correct, and do not produce changes to local load paths due to erroneous or undesired stiffness. To verify the relationship between degrees of freedom you may to use the menu:

- Patran: "Finite Element/Show/MPC" and "Utilities/Display/Show MPCs"

- Hypermesh: “Tool – Mask elements – by config” and “Tool – check elems – 1-d”.

7.3.1.10 Reference and Analysis Coordinate Systems

The orientation and location of all general and local co-ordinate systems must be confirmed. Plots must be made showing all model co-ordinate systems used.

7.3.1.11 Element connectivity

Plots showing element connectivity must be obtained to verify they are correct. This verification is very important to ensure the correct interpretation of MSC.Nastran element forces and the correct application of pressure loads. Correct connectivity is also required to guarantee the correct operation of any software used for automated strength analysis. The command to check connectivity for 2D (TRIA3 & QUAD4) elements in Patran is "Utilities/FEM Elements/Modify 2D Connectivity (Display)". This instruction displays a vector between the first and second node for each element. The command to display connectivity for 3D (Solid elements) is "Verify/Element/Connectivity" and in Hypermesh is “Tool – check elems – 2-d/3-d – connectivity”. This instruction will mark those elements with negative volume or questionable connectivity.

7.3.1.12 Element Offsets

Figures showing the element offsets in BAR/BEAM and QUAD4/TRIA3 must be obtained to verify the correct application in the model.

In Patran, to display the BAR/BEAM with offset, the function "Display/LBC/Element Properties/Beam Display 1D Line + Offset" option must be used. To display the offset in nodes as a vector, the function "Properties/Show/Offset @ Node 1-2" is used. To display the offset in plate elements, the function "Utilities/Display/Display Plate Offset" is used. The offsets values of concentrated masses (CONM2 cards) are obtained with the function "Properties/Mass Offset". The CONM2 cards in Patran are shown as a triangle located in the assigned node. There is, however, a reported error in Patran where the position of mass is defined in absolute co-ordinates (option CID = -1 in CONM2 card). In this situation Patran takes these co-ordinates as the offset values.

In Hypermesh, the visualization toolbar provides the tools required to plot the offsets and thicknesses of the 1-d and 2-d elements.

7.3.1.13 Element Co-Ordinate Systems

For the correct interpretation of element forces, stresses, and strains it is necessary to know the direction of the element co-ordinate system axes. With the function "Utilities/Display/Finite Element Coordinate Frames" in Patran you may be able to display the element co-ordinate systems for selected elements (select MSC.Nastran option).

Plots showing the element co-ordinate system for areas where consistent element orientations are necessary should be obtained, checking the axes in 1D (BAR/BEAM) and 2D (QUAD4/TRIA3) elements as necessary.

In Hypermesh a very detailed toolset is available in the Aerospace user profile: “Menu Bar: Aerospace Menu - Display - Orientation Review”

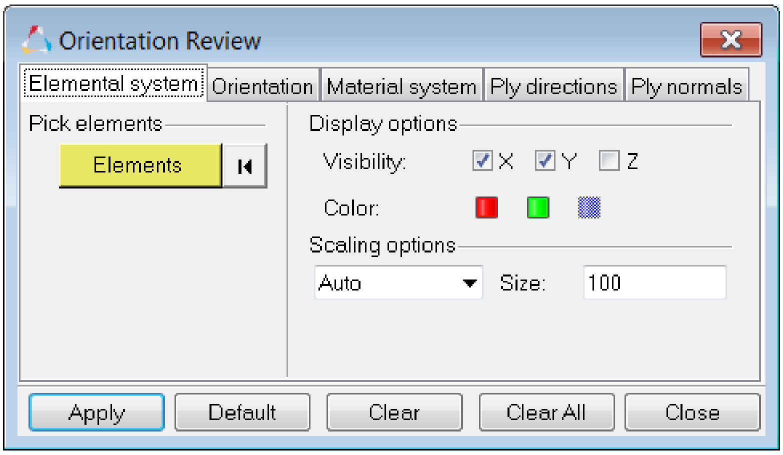


Fig. 7.6. Hypermesh. Aerospace Orientation review – Elemental system

7.3.1.14 Material Co-Ordinate Systems

When a model uses composite materials, the material co-ordinate system, and the relative angle between the material and the element co-ordinate systems must be verified and plotted.

It is recommended that the material axes are defined using an auxiliary global co-ordinate system in order to avoid problems due to irregular elements or future mesh modifications. Do not define Xelement by specifying the angle between G1G2 and the material axis in CQUAD4/CTRIA3 cards.

In Patran, the projection of the material co-ordinate system over the elements can be obtained with the command "Utilities/Display/Plot Material Orientation".

In Hypermesh these checks are also available in the Aerospace user profile: “Menu Bar: Aerospace Menu - Display - Orientation Review – Material system”

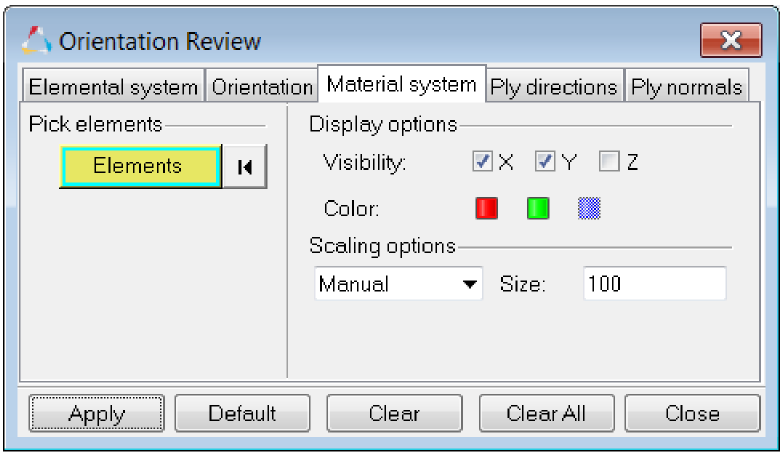


Fig.7.9. Hypermesh. Aerospace Orientation review – Material system

7.3.2 Geometrical Checks with MSC.Nastran

These checks are complementary to those performed with Pre-Processing Tool. Having verified the topology and properties, it is necessary to run an analysis of the model using MSC.Nastran to obtain one additional check for model consistency, and thus prevent a MSC.Nastran fatal error. It is not necessary to complete a full analysis run for this check, the analysis may be terminated after the data-checking phase of the analysis.

In MSC.Nastran the model must be executed with SOL 101 to verify the model geometry, and to calculate the load resultant for each of the verification static load cases.

MSC.Nastran will perform some element geometrical and connectivity checks. These are similar to the checks performed by Patran, but some of the assessment criteria used by MSC.Nastran differ from those used by Patran. It is still considered necessary to perform both sets of tests as the initial Patran checks will detect the most severe errors, leaving the MSC.Nastran checks to identify any residual problems. Verifications for TRIA3 & QUAD4 "Skew Angle" and for QUAD4 "Taper" are performed. The elements that do not pass the test are listed in the F06 file as either "User Information Message (UIM)" or "User Warning Message (UWM)". The "Skew Angle" will produce a warning message when the value is smaller than 30.0 degrees for QUAD4 elements, and 10 degrees for TRIA3 elements. "Taper" gives a warning message when is higher than 0.5. Both "Taper" and "Skew" are very important for the efficiency of the MSC.Nastran QUAD4 element, while large warping angles will create artificial local internal forces.

In order to perform the analysis check with MSC.Nastran it is necessary to select the following PARAM values:

a) PARAM, CHECKOUT, YES or CHECKOUT Executive Command to check the model geometry and the matrix consistency, and to produce the load resultant for the applied loads, in the origin of basic co-ordinate system (default). If you need the load resultant relative to a different point use the following parameter.

b) PARAM, GRPNT, G0 this defines the node about which to calculate the load resultant, the reactions, the mass and inertia, etc. If you do not use this parameter the default value is the origin of basic co-ordinate system.

c) PARAM, PRTMAXIM, YES To show the maximum displacements, maximum spcforces, maximum applied load and maximum mpcforces for each load case in basic co-ordinate system.

The MSC.Nastran execution shows also if there are RBEs not correctly defined or are quasi singular.

7.4 Guidelines for Static Analysis Checks

7.4.1 Constrains and Boundary Conditions

Constraint data and Boundary Conditions must be applied to the correct nodes/degrees-of-freedom, in the correct directions, and with the correct stiffness where relevant. Evidence must be produced to demonstrate that these criteria have been met.

Cases representing each of the maximum predicted loading cases should be used to evaluate the reaction distribution through each of the constraint systems, e.g. in a HTP FEM, the most critical bending and torsion load case should be used, and also the elevator Maximum Hinge Moment load case if possible, and the results of these checks must be included in Form 4.

The use of AUTOSPC = YES or PARAM, AUTOSPC, YES is only permitted during preparation and model verification model phase, and not in the final model. Any additional SPC's identified by AUTOSPC must produce negligible reactions within the final model. If these reactions are not negligible, then corrective action to the model is required.

SPC cards created automatically by MSC.Nastran creates when the model is run using the AUTOSPC(PUNCH, SID=10)=YES or PARAM AUTOSPC YES may be created automatically by executing the analysis using PARAM, SPCGEN, 1. This command produces a PUNCH file of the additional SPC cards identified by AUTOSPC to constraint the model accordingly. These cards can then be included in the Bulk Data of the FEM for the final check runs using AUTOSPC = NO or PARAM AUTOSPC NO.

N.B.: Except when part of the main support of the model, it is only acceptable to SPC rotational DOFs, never translation DOFs, translation constraints must be justified even when they generate zero reactions.

This restriction also applies for dynamic verifications, as it is the intention to use the same model for both static and dynamic calculations. This will therefore be the normal configuration.

A general sketch/plot of each set of boundary conditions used in the model must be created for the purposes of verification and documentation.

The reaction resultant, i.e. SPCFORCE RESULTANT, at the reference point must be equal and opposite in sign to the applied load resultant, i.e. OLOAD RESULTANT.

7.4.2 Applied Loads

It must be confirmed that all loads applied to the model are applied in the correct directions, have the correct magnitudes, and are at the correct nodal locations.

The following specific items must be taken into account:

- The total applied load in all 6 DOFs with respect to the reference point. This point is by default the origin of the basic co-ordinate system.
- The loading distributions along the component main dimension for shear load, bending moment, and torque moment derived using MSC.Nastran internal loads.

- The comparison between applied and theoretical load distributions from MSC.Nastran, at each load monitoring station (dbf files), e.g. ribs or frame stations.
- The reaction resultant (SPCFORCE RESULTANT) is equal and opposite of the applied load resultant (OLOAD RESULTANT). This output is printed automatically (F06 file) in solutions such as SOL 101.

These checks will be included in a technical report and referenced on Form 4.

When it is necessary to introduce new nodes for loading introduction, it is recommended that they are included by using an RBE3. This RBE3 is defined as follows:

The new node must be the "REFERENCE GRID" for the RBE3 with either 6 dependent degrees-of-freedom, unless other specific configuration is required according to Stress Methods and Optimization rules.

7.5 Guidelines for Dynamic Verification

Performing a normal modes calculation allows confirmation that all the nodes are adequately connected to the structure, and there are no mechanisms or low stiffness areas.

This check is made using SOL 103 of MSC.Nastran using the LANCZOS method.

As mentioned in section 3.6, dynamic verification shall be performed in two phases:

- With densities in all Material Cards. The object of these runs are:
  - GFEM modal verification: detect and correct undesired local modes (see Fig. 7.10)
  - Perform Rigid body checks with GROUNDCHECK
- Removing the densities from the Material cards and adding the component weight by means of CONM2, according to weight distributions provided by Weight and Balance Team. The objectives of these runs are:
  - Determination of the flexible modes of the component

<u>REMARKS</u>

a) All analyses must be made with AUTOSPC = NO or “PARAM, AUTOSPC, NO”. This is normal practice.

b) Translation singularities must be eliminated with the correct use of RBE elements. Any rotational singularities not constrained by using PARAM, K6ROT, 1. And PARAM, SNORM, 20. Must be eliminated with SPC’s or RBE’s, as normal practice.

c) The use of field 8 of the GRID card or GRIDSET cards to include permanent single point constraints is forbidden. This is normal practice as the use of that field can cause the inadvertent grounding of masses during dynamic analyses or loads during static analyses if the node is used as a mass or loading point.. Singularities should be eliminated via selected and verified SPC1 or SPC cards.

d) The deflected shape of the model should be assessed, with drawings of the deformed structure, using the basic co-ordinate system. Areas where the displacement co-ordinate system is not the basic one should be specifically identified and plotted where possible.

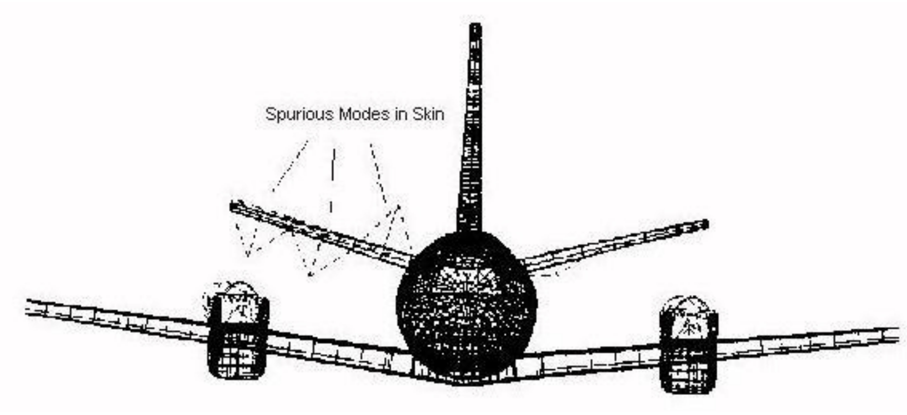


Fig. 7.10. Local or spurious mode in HTP Skin

Rigid Body Check

The Rigid Body Check shall be performed during the Check 1 described in 3.6.

The GROUNDCHECK Nastran Card (Case Control Section) shall be used to identify unintentional constraints and ill conditioning in the stiffness matrix.

GROUNDCHECK(SET=ALL,GRID=ngrid,DATAREC=YES,RTHRESH=0.1)=YES

will request a grounding check of all DOF sets:

• G g-set before single point, multipoint constraints, and rigid elements are applied

• N n-set after multipoint constraints and rigid elements are applied

• F f-set after single point, multipoint constraints, and rigid elements are applied

• A a-set after static condensation

ngrid must be chosen near the Center of Gravity of the component model

This check could be replaced by enforced displacement test but groundcheck is the preferred option (more powerful in order to detect the problems).

7.6 Guidelines for Thermoelastic Verification

To satisfy this "Zero Stress" check requested in paragraph 3.7, a mechanically verified model must be modified, according to the following rules:

Elements:

- CROD: No Restriction
- CBAR - CBEAM:

− The offsets of shear axis must not modify the distance between the 2 nodes of the beam (vectors W1 and W2).

− The offsets of neutral axis (BEAM) have no influence.

- CTRIA3-CQUAD4 :

  − If the offsets ZOFFS or Z0 are used, the parameter SNORM must be set to 0.5 instead of 20 (due to Nastran limitations)

  − A warped QUAD results in a non-zero stress (Nastran limitation) but is generally acceptable (under the accuracy level)

- CTETRA-CHEXA: No Restriction
- GENEL-DMIG: Generally non valid Elements for a thermo-elastic analysis. (Except under circumstances where their inclusion does not modify the displacements in another part of the model)
- CELASx/CBUSH:
  - CELASx/CBUSH between two coincident nodes: No Restriction
  - CELASx/CBUSH of non-zero length not valid for this check: see GENEL-DMIG above.

D.O.F RELATIONS:

- RBE2/RBAR:
  - RBE2/RBAR between two coincident nodes for 6dof: No Restriction
  - Otherwise:
    - Nastran version 2005 and beyond: specify RIGID=LAGR in Case Control Section and introduce ALPHA value (the same used for the rest of MAT cards) in RBE2/RBAR card, or
    - see GENEL-DMIG
- RBE3:
  - If the slave node has no associated rigidity (Mass or Load introduction node): No Restriction
  - Otherwise:
    - Nastran version 2005 and beyond: specify RIGID=LAGR in Case Control Section and introduce ALPHA value (the same used for the rest of MAT cards) in RBE3 card, or
    - see GENEL-DMIG
- MPC:
  - MPC linking coincident nodes: No Restriction
  - If the slave node is a barycenter of two master nodes with coefficients in the same ratio as the distances: No Restriction
  - Otherwise: see GENEL-DMIG

7.7 How to Fulfill the Verification Forms

The following pages contain explanatory guidelines to fulfill the Forms required for conformance to this GFEM Verification procedure.

Two sets of guidelines are presented; the first contains explanatory text defining the contents of sections of the forms to assist the user in completing them.

The second set contains a complete example based on some Airbus DS real examples.

For the convenience of the user, the two sets have been interleaved so that the explanation guidelines for each form are immediately followed by the example for that form.

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION **REFERENCES**	**PROJECT:** <A/C or project>
PART COMPONENT:	**A/C COMPONENT:** <Full A/C or component identification>	**PAGE:** 1
GFEM IDENTIFICATION: < 16 alphanumeric code identifying GFEM>		**VERSION:** <two digits for version>

Finite Element Model general description:

< Analyst should provide a simple description of the FE model, if necessary outlining the function to which it has been developed >

Reference Coordinate System (FEM Component related to DBD/CATIA)

< A definition of the geometric axis system used for the model data, and the relationship of this system to that given within the relevant DBD. Include a reference for the source of data used. >

Geometry

< Reference to a report or other form of data source, e.g. a CADs database or list of drawing references, used to define the geometry data used within the model. If necessary append a separate list to the end of this report and refer to the list here. >

Meshing

< Reference to a report (DPI-ADC-006) or other type of data source used to define the standard of meshing density used within the model. >

Properties

< Reference to a memorandum, report or other form of data source used to define the standard of idealisation used within the model. If necessary append a separate list to the end of this report and refer to the list here. >

Materials

< Reference to a report or other form of data source used to define the materials used within the model, and the data values assigned to them. If necessary append a separate list to the end of this report and refer to the list here. >

	Name (Dept.)	**Signature**	**Date**
Compiled by:	<author>		
Approved by:	<GFEM authority or Stress Methods>		

DPI-ADC-007 Issue F – Form 1

AIRBUS DEFENCE & SPACE	**GFEM VERIFICATION** **REFERENCES**	**PROJECT:** C295MW
PART COMPONENT: Primary Structure	**A/C COMPONENT:** Full A/C	**PAGE:** 1
GFEM IDENTIFICATION: C295ETOTWF00S	**VERSION:** -01	

Finite Element Model general description:

C295MW GFEM. Flap 00 configuration
GFEM for internal Loads Analysis.
Supporting Stress and Fatigue Analysis
DOF = 38062
Description in ME-5-NT-140094-A

Reference Coordinate System (FEM Component related to DBD/CATIA)

C295 GFEM reference is shifted related DMU/CATIA Reference system
X = -1524. mm
Y = 0.
Z = -3000. mm

Geometry

Reference Drawing: 95-00048-00

Meshing

According with DPI-ADC-006 and C295 GFEM Reference DT-5-ADS-00017-A

Properties

C295 GFEM Properties defined as per DT-5-ADS-00017-A.
C295MW modified properties defined in ME-5-NT-140094-A

Materials

C295 GFEM Material Properties defined as per DT-5-ADS-00017-A.
C295MW GFEM modified area defined in ME-5-NT-140094-A
Reference data for Material Props: DAMA 8.3

	Name (Dept.)	Signature	Date
Compiled by:	S.P.M. (Empresa X)		
Approved by:	Resp. GFEM		

DPI-ADC-007 Issue F – Form 1

AIRBUS DEFENCE & SPACE	**GFEM VERIFICATION** **FILES RECORD**	**PROJECT:** <A/C or project>
PART COMPONENT:	**A/C COMPONENT:**	**PAGE:** 2
GFEM IDENTIFICATION: < 16 alphanumeric code identifying GFEM>		**VERSION:** <two digits for version>

COMPUTER DATASET / FILE NAME	**DATE OF CREATION**	**COMMENTS/DESCRIPTION**
<Enter list of ALL associated computer files for this model and the verification activity. - Bulk Data - Macros - Test files - MSC/Nastran run outputs - Patran/Hyperworks post-processing files>		<Provide a short description for each file or groups of files>

	Name (Dept.)	**Signature**	**Date**
Compiled by:	<author>		
Approved by:	<GFEM authority or Stress Methods >		

DPI-ADC-007 Issue F – Form 2

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION **FILES RECORD**	**PROJECT:** C295MW
PART COMPONENT: Primary Structure	**A/C COMPONENT:** Full A/C	**PAGE:** 2
GFEM IDENTIFICATION: C295ETOTWF00S	**VERSION:** -01	

COMPUTER DATASET / FILE NAME	DATE OF CREATION	COMMENTS/DESCRIPTION
/home/calculo-fem/c295/W/S1/FEM/		**GFEM Folder**
/FUS/ C295W_FEM-LIM_F00.dat	04/09/13	Nodos “dummy” para F00. Con Winglet.
/FUS/ C295W_FEM-LIM_F00_sW.dat	18/06/13	Nodos “dummy” para F00. Con C295 wingtip
/FUS/ C295W_FEM-LIM_F23.dat	04/09/13	Nodos “dummy” para F23. Con Winglet.
/FUS/ C295W_FEM-LIM_F23_sW.dat	03/09/13	Nodos “dummy” para F23. Con C295 wingtip
/FUS/ C295W_FUS_CEN.dat	14/01/13	Fuselaje: Secciones Fr. 21 a 24
/FUS/ C295W_FUS_CFWD.dat	14/01/13	Fuselaje: Secciones Fr. 11 a 10
/FUS/ C295W_FUS_COLIN.dat	09/01/13	Fuselaje: Secciones Fr. 47 a 52
/FUS/ C295W_FUS_CREB.dat	14/01/13	Fuselaje: Secciones Fr. 24.3 a 25
/FUS/ C295W_FUS_CRWD.dat	14/01/13	Fuselaje: Secciones Fr. 25 a 30
/FUS/ C295W_FUS_NOSE.dat	09/01/13	Fuselaje: Secciones Fr. 1 a 10
/FUS/ C295W_FUS_REAR.dat	09/01/13	Fuselaje: Secciones Fr. 31 a 46
/GEN/ C295W_MAT_PROP.dat	11/09/13	Propiedades y materiales con densidades
/GEN/ C295W_MAT_PROP_noRO.dat	11/09/13	Propiedades y materiales sin densidades
/GEN/ C295W_SPC_MPC_CORD_ROT.dat	09/01/13	Condiciones de contorno: SPCs y MPCs.
/GEN/ C295W_SPC_SING_F00.dat	16/11/15	Tarjetas SPC para Flap 0º
/GEN/ C295W_SPC_SING_F23.dat	16/11/15	Tarjetas SPC para Flap 23º
/MASS/ _C295W_EOW_CONM2.dat	25/06/13	Tarjetas CONM2 para configuración EOW.
/OPT/ C295_DYN_AERO.dat	05/09/13	Conecta paneles aerodinámicos al FEM
/TP/ C295W_ELEV_RUD.dat	18/06/13	Elevator y Rudder
/TP/ C295W_HTP.dat	18/06/13	Estructura del HTP
/TP/ C295W_VTP.dat	18/06/13	Estructura del VTP
/WING/ C295W_AIL.dat	18/06/13	Alerones
/WING/ C295W_EMS.dat	18/06/13	Bancada motor
/WING/ C295W_FLAP00.dat	18/06/13	Flaps 0º
/WING/ C295W_FLAP23.dat	02/07/09	Flaps 23º
/WING/ C295W_TAB.dat	25/06/13	Tabs
/WING/ C295W_WBOX_S1.dat	04/09/13	Cajón de torsión del ala
/WING/ C295W_WF_INTF.dat	18/06/13	Elementos de interfaz ala-fuselaje
/WING/ C295W_Winglet_35kg.dat	25/06/13	Modelo de barras del Winglet
/home/calculo-fem/c295/W/S1/Valid_C295WF00S01/		
C295W_S1_F00_chk_rbody1.dat		**Verification Folder**
C295W_S1_F00_modal1.dat	12/11/15	Chequeo de Sólido Rígido
C295W_S1_F00_modal2.dat	12/11/15	Modos Normales: Pasada 1
C295W_S1_F00_modal3.dat	12/11/15	Modos Normales sin actuadores.: Pasada 2
C295W_S1_F00_modal4.dat	12/11/15	Modos Normales: Pasada 3
C295W_S1_F00_Static_Val.dat	12/11/15	Modos Normales sin uniones: Pasada 4
	12/11/15	Pasada de validación estática

	Name (Dept.)	Signature	Date
Compiled by:	S. Pérez (CT Ingenieros)		
Approved by:	Resp. GFEM		

DPI-ADC-007 Issue F – Form 2

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION **MODEL CHECK LIST**	**PROJECT:**

PART COMPONENT:	**A/C COMPONENT:**	**PAGE:** 3

GFEM IDENTIFICATION: < 16 alphanumeric code identifying GFEM>	**VERSION:** <two digits for version>

MODEL SIZE

GRIDs	1-D	2-D	3-D	0-D	MPC / RBE
<nb of grids>	<nb of 1d el>	<nb of 1d el>	<nb of 1d el>	<nb of conm2/elas>	<nb of mpcs & RBEs>

CHECK LIST	DEPARTMENT RESPONSIBLE & PRINCIPAL DATA VALUE (If it is relevant)	SIGNATURE
FEM Identification Criteria	<verification of the ID ranges for elem and nodes>	
Coincident Nodes		
Coincident Elements		
Free Edges/Free Faces		
Element Warping		
Element Twist		
Interior Angles		
Distortion of Elements		
Connectivity		
Bending Planes (BAR&BEAMs)		
Offsets (all elements)		
Positive normal (QUAD4/TRIA3)		
Element Co-ordinate System		
Material Co-ordinate System		
Co-ordinate Systems		
Rigid Elements, MPCs		
Degrees of Freedom (KLL – F04)	<identify DOFs from KLL dimensions in f04 output file>	

ANALYSIS DATA

Computer	Operating System	MSC.Nastran Version
<Model>	<extract from f06>	<version and release id>

Max. Diagonal Factor	<maxratio from f06>	**Is Grid Point Singularity. Table Empty?**

If YES		**If NO**	**Attached as page No.:**	<page with singular.>

MSC.NASTRAN Parameters
< Identify NASTRAN param used in the GFEM run>

COMMENTS:
< Against each entry in the table above, enter Name and Department/Group reference or Siglum of the person responsible for completing the checklist item. Also, where relevant, give the most appropriate value, e.g. for Element Twist quote the worst twist measured. If required, hard copy output can be appended and a reference made to the relevant page numbers in this report. The person named should then sign the end column. >
< For the rest of the form, complete the entries and sign below as Compiler. >

	Name (Dept.)	Signature	Date
Compiled by:			

Approved by:			

DPI-ADC-007 Issue F – Form 3

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION **MODEL CHECK LIST**	**PROJECT:**

PART COMPONENT: Primary Structure	**A/C COMPONENT:** Full A/C	**PAGE:** 3
GFEM IDENTIFICATION: C295ETOTWF00S	**VERSION:** -01	

MODEL SIZE

GRIDs	1-D	2-D	3-D	0-D	MPC / RBE
14522	16687	10852	0	0	1518

CHECK LIST	DEPARTMENT RESPONSIBLE & PRINCIPAL DATA VALUE (If it is relevant)	SIGNATURE
FEM Identification Criteria	OK. MSC.PATRAN V11.0	**SPM**
Coincident Nodes	OK. MSC.PATRAN V11.0	**SPM**
Coincident Elements	OK. MSC.PATRAN V11.0	**SPM**
Free Edges/Free Faces	OK. MSC.PATRAN V11.0	**SPM**
Element Warping	OK. PATRAN V11.0 / MSC.NASTRAN V2005.5.2	**SPM**
Element Twist	OK. MSC.PATRAN V11.0	**SPM**
Interior Angles	OK. PATRAN V11.0 / MSC.NASTRAN V2005.5.2	**SPM**
Distortion of Elements	OK. PATRAN V11.0 / MSC.NASTRAN V2005.5.2	**SPM**
Connectivity	OK. MSC.PATRAN V11.0	**SPM**
Bending Planes (BAR&BEAMs)	OK. MSC.PATRAN V11.0	**SPM**
Offsets (all elements)	OK. MSC.PATRAN V11.0	**SPM**
Positive normal (QUAD4/TRIA3)	OK. MSC.PATRAN V11.0	**SPM**
Element Co-ordinate System	OK. MSC.PATRAN V11.0	**SPM**
Material Co-ordinate System	OK. MSC.PATRAN V11.0	**SPM**
Co-ordinate Systems	OK. MSC.PATRAN V11.0	**SPM**
Rigid Elements, MPCs	OK. MSC.NASTRAN V2005.5.2	**SPM**
Degrees of Freedom (KLL – F04)	38062	**SPM**

ANALYSIS DATA

Computer	Operating System	MSC.Nastran Version
Multivac	Sun Os 5.10	2005r3b (2005.5.2)

Max. Diagonal Factor	2.94E+05	**Is Grid Point Singularity. Table Empty?**			
If YES	**YES**	**If NO**	**Attached as page No.:**	**N/A**	N/A

MSC.NASTRAN Parameters
NASTRAN K6ROTMEM=1, OLDQ4K=2, OLDRBE3=1
PARAM,SNORM,20. ; PARAM,K6ROT,1.

COMMENTS:

No Tabs have been included for FEM Verification.

Singularities introduced with SPC Cards

	Name (Dept.)	Signature	Date
Compiled by:	S. Pérez (CT Ingenieros)		

Approved by:	Resp. GFEM		

DPI-ADC-007 Issue F – Form 3

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION **STATIC ANALYSIS CHECK LIST**	**PROJECT:**

PART COMPONENT:	**A/C COMPONENT:**	**PAGE:** 4

GFEM IDENTIFICATION: < 16 alphanumeric code identifying GFEM>	**VERSION:** <two digits for version>

BOUNDARY CONDITIONS DESCRIPTION:
(text, figure or references)

<Description of the boundary conditions. Grids and dofs supported or plot or reference to document>

LOAD CASE DESCRIPTION:

<Short description of the Load Case used for check: Limit Load (Load Case ID), Book case, max bending, fuselage pressure.>
<call to image or document if needed>

Load Balance Checks	**FX**	**FY**	**FZ**	**MX**	**MY**	**MZ**
Applied	<OLOAD	Resultant	Extracted	From F06	File>	
Reaction (SPCs)	<SPC	FORCE	Resultant	From F06	File>	
Difference (%)						
Max. Displacements	**TX**	**TY**	**TZ**	**RX**	**RY**	**RZ**
GRID	(*)					
Value	(*)					
Max. SPCFORCES	**FX**	**FY**	**FZ**	**MX**	**MY**	**MZ**
GRID	(**)					
Value	(**)					
Epsilon	<Value from F06>		**External Work**		<Value from F06>	

COMMENTS:
(*) Grid that has the maximum displacement or rotation at each direction. One grid per direction

(**) GRID that has the maximum SPCFORCE at each direction

<Insert as explanations needed, NASTRAN parameters, identification of differences, etc.>

	Name (Dept.)	**Signature**	**Date**
Compiled by:			

Approved by:			

DPI-ADC-007 Issue F – Form 4

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION **STATIC ANALYSIS CHECK LIST**	**PROJECT:** C295 SA04
PART COMPONENT: Primary Structure	**A/C COMPONENT:** Full A/C	**PAGE:** 4
GFEM IDENTIFICATION: C295ETOTWF00S	**VERSION:** -01	

BOUNDARY CONDITIONS DESCRIPTION:
(text, figure or references)

The GFEM is supported in the Wing-to-Fuselage interface points, as described in C295 GFEM Reference DT-5-ADS-00017-A

LOAD CASE DESCRIPTION:

LC1=LC 100000; Fuselage Pressure: 5.77 psi

Nodal Loads. From C295 LIM, see Ref.XX

Load Balance Checks	**FX**	**FY**	**FZ**	**MX**	**MY**	**MZ**
Applied	4.282	-11.047	31.072	3.78E+04	1.41E+05	5.50E+04
Reaction (SPCs)	-4.282	11.036	-30.775	-3.77E+04	-3.67E+04	-3.46E+04
Difference (%)	0.00	0.10	0.96	0.22	73.92	37.18
Max. Displacements	**TX**	**TY**	**TZ**	**RX**	**RY**	**RZ**
GRID	6904503	164214	152099	167803	103211	129216
Value	18.85	10.95	23.04	0.10	0.36	0.16
Max. SPCFORCES	**FX**	**FY**	**FZ**	**MX**	**MY**	**MZ**
GRID	121991	121991	121991	130224	115290	115286
Value	4.28	26.09	42.96	45.86	370.19	6026.98
Epsilon	-8.6688089E-13		**External Work**		6.4839505E+06	

COMMENTS:

NASTRAN PARAMs
PARAM,POST,-1; AUTOSPC,NO; NOCOMPS,-1; GRDPNT,9999; PRTMAXIM,YES; WTMASS,1.E-3: Densities in Kg/mm3

	Name (Dept.)	**Signature**	**Date**
Compiled by:	S.P.M. (Empresa X)	x	x

Approved by:	Resp. GFEM	x	x

DPI-ADC-007 Issue F – Form 4

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION **ANALYSIS CHECK LIST: NORMAL MODES 1**			**PROJECT:**
PART COMPONENT:		**A/C COMPONENT:**		**PAGE:** 5
GFEM IDENTIFICATION: < 16 alphanumeric code identifying GFEM>			**VERSION:** <two digits for version>	

	G-Set	N-Set	F-Set	A-Set
Rigid Body Check	<ok/not ok>	<ok/not ok>	<ok/not ok>	<ok/not ok>
	Check 1	**Check 2**	**Check 3**	**Check 4**
Mass Reference	<Ref. or estimated	Mass>		
Mass Output from GPWG	< f06 extracted>			
Mode Plots & Frequency List	<Ref>			

	Frequency [Hz]			
	Check 1	**Check 2**	**Check 3**	**Check 4**
Mode No:	(Densities in MATx) [-1,1] free/free with actuators	(Densities in MATx) [0,1] free/free without actuators	(Densities in MATx) [1,∞] free/free with actuators	(Densities in MATx) [0,∞] free/free with interfaces disconnected
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				

Comments:

< From the normal modes analysis, complete the table with F06 values.
Only entries in the UNSHADED areas will be used for the assessment. >

<Comments: identify local modes or unexpected values. PARAM WTMASS value, etc>

< NB. This sheet requires a DYNAMIC acceptance signature in addition to the normal approval. >

	Name (Dept.)	Signature	Date
Compiled by:			
Approved by:			

Dynamic Acceptance:			

DPI-ADC-007 Issue F – Form 5

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION ANALYSIS CHECK LIST: NORMAL MODES 1	PROJECT: C295 MW

PART COMPONENT: Primary Structure	A/C COMPONENT: Full A/C	PAGE: 5
GFEM IDENTIFICATION: C295ETOTWF00S	VERSION: -01	

	G-Set	N-Set	F-Set	A-Set
Rigid Body Check	OK	OK	OK	OK
	Check 1	**Check 2**	**Check 3**	**Check 4**
Mass Reference	9.523E+03	9.523E+03	9.523E+03	9.523E+03
Mass Output from GPWG	3.506520E+03	3.506520E+03	3.506520E+03	3.506520E+03
Mode Plots & Frequency List	ME-5-NT-140094	ME-5-NT-140094	ME-5-NT-140094	ME-5-NT-140094

	Frequency [Hz]			
	Check 1	**Check 2**	**Check 3**	**Check 4**
Mode No:	(Densities in MATx) [-1,1] free/free with actuators	(Densities in MATx) [0,1] free/free without actuators	(Densities in MATx) [1,∞] free/free with actuators	(Densities in MATx) [0,∞] free/free with interfaces disconnected
1	1.221084E-05	3.542312E-05		2.159260E-05
2	1.275197E-04	1.355410E-04		1.212106E-04
3	1.688949E-04	1.637824E-04		1.634745E-04
4	6.961310E-02	6.961112E-02		6.961112E-02
5	7.035195E-02	7.034785E-02		7.034788E-02
6	1.506940E-01	1.506142E-01		1.506142E-01
7		3.552366E-01	5.532002E+00	3.552366E-01
8		8.893999E-01	6.205381E+00	8.893999E-01
9		8.993931E-01	7.532038E+00	8.993931E-01
10			9.708539E+00	5.746063E+00
11			1.002005E+01	6.230577E+00
12			1.136154E+01	7.542253E+00
13			1.293248E+01	9.719954E+00
14			1.450627E+01	1.020197E+01
15			1.509752E+01	1.153385E+01

Comments:

Rigid Body Checks: Grouncheck Command: See Table Pag. XX
Movable Surfaces considered for Check 2: 3 (Rudder and Elevators)
NASTRAN PARAMS
AUTOSPC,NO; PRTMAXIM,YES; POST,-1; WTMASS, 1.0E-03; K6ROT,1.0; SNORM,20.

	Name (Dept.)	Signature	Date
Compiled by:	S.P.M. (Empresa X)		
Approved by:	Resp. GFEM		

Dynamic Acceptance:	Resp. DYN		

DPI-ADC-007 Issue F – Form 5

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION **ANALYSIS CHECK LIST: NORMAL MODES 2**	**PROJECT:**

PART COMPONENT:	**A/C COMPONENT:**	**PAGE:** 6
GFEM IDENTIFICATION: < 16 alphanumeric code identifying GFEM>	**VERSION:** <two digits for version>	

BOUNDARY CONDITIONS DESCRIPTION:
(text, figure or references) < Specific Boundary conditions description for normal modes. Default: free-free>

	Check 5	Check 6	Check 7	Check 8
Mass Reference	<Ref or estimated mass>	<Ref or estimated mass>	<Ref or estimated mass>	<Ref or estimated mass>
Mass Output from GPWG	< f06 extracted>	< f06 extracted>	< f06 extracted>	< f06 extracted>
Mode Plots & Frequency List	<Ref>			

	Frequency [Hz]			
	Check 5 (M1)	**Check 6 (M2)**	**Check 7 (M2)**	**Check 8 (M2)**
Mode No:	(Masses in CONMx) [-1,1] free/free with actuators	(Masses in CONMx) [0,∞] free/free without actuators	(Masses in CONMx) [1,∞] free/free with actuators	(Masses in CONMx) [0,∞] free/free with interfaces disconnected
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				

Comments:
< From the normal modes analysis, complete the table with F06 values.
Only entries in the UNSHADED areas will be used for the assessment. >

<Comments: identify local modes or unexpected values. PARAM WTMASS value, etc>

< NB. This sheet requires a DYNAMIC acceptance signature in addition to the normal approval. >

	Name (Dept.)	Signature	Date
Compiled by:			
Approved by:			
Dynamic Acceptance:			

	GFEM VERIFICATION **ANALYSIS CHECK LIST: NORMAL MODES 2**	**PROJECT:** C295 MW

PART COMPONENT: Primary Structure	**A/C COMPONENT:** Full A/C	**PAGE:** 6
GFEM IDENTIFICATION: C295ETOTWF00S	**VERSION:** -01	
BOUNDARY CONDITIONS DESCRIPTION: free-free condition		

	Check 5	**Check 6**	**Check 7**	**Check 8**
Mass Reference	9.523E+03	N/A	9.523E+03	N/A
Mass Output from GPWG	9.3675+03	N/A	9.3675+03	N/A
Mode Plots & Frequency List	DT-5-ADS-00017-A	DT-5-ADS-00017-A	DT-5-ADS-00017-A	DT-5-ADS-00017-A

	Frequency [Hz]			
	Check 5 (M1)	**Check 6 (M2)**	**Check 7 (M2)**	**Check 8 (M2)**
Mode No:	(Masses in CONMx) [-1,1] free/free with actuators	(Masses in CONMx) [0,∞] free/free without actuators	(Masses in CONMx) [1,∞] free/free with actuators	(Masses in CONMx) [0,∞] free/free with interfaces disconnected
1	1.21E-04		1.28E+01	
2	8.10E-05		1.36E+01	
3	4.28E-05		1.65E+01	
4	6.02E-05		1.69E+01	
5	7.78E-05		1.81E+01	
6	1.74E-04		2.34E+01	
7			2.49E+01	
8			2.58E+01	
9			2.64E+01	
10			3.27E+01	
11			3.49E+01	
12			3.64E+01	
13			3.96E+01	
14				
15				

Comments:

NASTRAN PARAMS
AUTOSPC,NO; PRTMAXIM,YES; POST,-1; WTMASS, 1.0E-03; K6ROT,1.0; SNORM,20.
CONM2 cards in Kg

Checks 6 and 8 agreed with Dynamic Team to skip because considered enough Checks 2 and 4

	Name (Dept.)	**Signature**	**Date**
Compiled by:	S.P.M. (Empresa X)		
Approved by:	Resp. GFEM		
Dynamic Acceptance:	Resp. DYN		

DPI-ADC-007 Issue F – Form 6

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION **THERMO-ELASTIC ANALYSIS CHECK LIST**	**PROJECT:**

PART COMPONENT:	**A/C COMPONENT:**	**PAGE:** 7

GFEM IDENTIFICATION: < 16 alphanumeric code identifying GFEM>	**VERSION:** <two digits for version>

BOUNDARY CONDITIONS DESCRIPTION:
(text, figure or references)

Isostatic constraints <Identify GRIDS and D.o.f.s>

LOAD CASE DESCRIPTION:

Defined Temperature : <Default: 0-100 C or 20-120 C>

Load Balance Checks	**FX**	**FY**	**FZ**	**MX**	**MY**	**MZ**
Applied						
Reaction (SPCs)						
Difference (%)						
Stress	**MAX**	**MIN**				
ELEMENT						
Value						
Epsilon			**External Work**			

MSC.NASTRAN Parameters
< AUTOSPC, K6ROT, RIGID=LAGRANGE, etc>

COMMENTS:
< Identify areas with out of tolerances or problematic points>

	Name (Dept.)	**Signature**	**Date**
Compiled by:			
Approved by:			

DPI-ADC-007 Issue F – Form 7

AIRBUS DEFENCE & SPACE	GFEM VERIFICATION **THERMO-ELASTIC ANALYSIS CHECK LIST**	**PROJECT:** A400M

PART COMPONENT: HTP	**A/C COMPONENT:** Full HTP & Elevators	**PAGE:** 7
GFEM IDENTIFICATION: A400MEHTPS96F	**VERSION:** -02	

BOUNDARY CONDITIONS DESCRIPTION:
(text, figure or references)
Isostatic constraints
Interfaces with VTP support
THSA Fitting (Grid 5700350)➔ Z constraint
Rear support Fittings (Grid 5506100 and 5706100) ➔ X-Z constraint
Lateral Bar (Grid 5700100) ➔ Y constraint

LOAD CASE DESCRIPTION:

Defined Temperature : $\Delta T=100^{o}$
(unitary thermal expansion coefficient for MATX entries: $\alpha=1.0\ 10^{-6}$)

Load Balance Checks	FX	FY	FZ	MX	MY	MZ
Applied	-2.11E-10	3.81E-10	1.45E-10	-7.85E-07	5.67E-06	1.07E-05
Reaction (SPCs)	9.39E-07	-2.77E-06	1.03E-06	2.54E-02	-3.80E-02	-1.25E-01
Difference (%)	low	low	low	low	low	low
Stress	**MAX**	**MIN**				
ELEMENT	58224508	58230508				
Value	1.47E-2	-1.48E-2				
Epsilon	-8.6691069E-11		**External Work**		6.0379E+04	

MSC.NASTRAN Parameters
AUTOSPC,NO; PRTMAXIM,YES; POST,-1; WTMASS, 1.0E-03; K6ROT,1.0; SNORM,20.

COMMENTS:
Results to be factorised by 23.4 to reach Aluminium thermal expansion coefficient.

The FEM is valid for thermal analysis except close to RBE with IDS (55824200 55824300 55828200 55828300 58830300)

	Name (Dept.)	Signature	Date
Compiled by:	S.P.M. (Empresa X)		
Approved by:	Resp. GFEM		

DPI-ADC-007 Issue F – Form 7

8 REFERENCED DOCUMENTS

Where a specific issue of the document is provided, it shall apply. When documents are referenced herein, a short form quoting only the basic number of the document is used.

8.1 AIRWORTHINESS

N/A	

8.2 INDUSTRY STANDARDS

ISO/IEC 2382:2015	Information Technology - Vocabulary
ASME V&V 10-2006	Guide for Verification and Validation in Computational Solid Mechanics
NAFEMS QSS (ISBN 978-1-874376-88-0)	Quality Management in Engineering Simulation: A primer for NAFEMS QSS

8.3 AIRBUS DEFENCE & SPACE DOCUMENTS

NT-T-ADP-10003-B	Guía para la realización de Análisis de Estructuras mediante Modelos de Elementos Finitos de Detalle

8.4 AIRBUS DEFENCE & SPACE PROCEDURES

DPI-ADC-006	Criterios Generales de Modelización por Elementos Finitos para MSC.Nastran
DP-ADC-002	Validación de resistencia estática de la estructura
DPI-ADC-009	Instrucción para la Realización de la Documentación Técnica del Departamento de Estática
DP-A00-008	Procedimiento de Archivo de Datos de Documentación
DP-000-049	Gestión electrónica de Documentos en la

	DPS

8.5 OTHER DOCUMENTS

M2036	Airbus FEA Manual
N/A	MSC.NASTRAN Quick Reference Guide

9 GLOSSARY

Assembly	A number of parts, subassemblies or any combination thereof joined together to perform a specific function and which can be disassembled without destruction of designed use
Component	Any self-contained part, combination of parts, subassemblies or units, which perform a distinctive function necessary to the operation of a system
Computer	See ISO 2382:2015
Data	See ISO 2382:2015
DFEM	Detailed Finite Element Model
File	See ISO 2382:2015
GFEM	Global Finite Element Model
Input (Bulk Data)	See ISO 2382:2015
LIM	Loads Interface Model: LIM is a Finite Element Model in NASTRAN Format containing the aerodynamic surface and inertial points, used to translate the Loads data into Forces and Moments in GFEM
Quality Assurance	All those planned or systematic actions necessary to provide adequate confidence that a product or service will satisfy given needs
Validation	The act of confirming that a specific requirement has been met. In simulation is referred to the accuracy of results
Verification	The act of confirming that a specific requirement has been met. In simulation is referred to inputs and process

10 REVISIONS RECORD

Edición *Issue*	**Fecha** *Date*	**Razones de los Cambios /** *Reasons for Revision-* ***Páginas/Secciones afectadas /*** ***Pages/Sections affected***	***Aprobación Propietario*** **Owner Approval**	***Autorización*** **Authorization**
A	Nov 1992	First Issue/All	A. González	I. A. Franganillo
B	Apr 1996		A. González	I. A. Franganillo
C	May 1999		A. González	M. A. Morell
D	Mar 2003	Complete Document Review	E. Oslé	J. M. Ruiz Barragán
E	Nov 2006	Improvement Actions (IA-0-ADC-007/05), updated Format. Document review	E. Oslé	J. L. Fauste
F	Dec 2016	New format translation into English, change of legal entity, complete Document Review	E. Oslé	E. Mennle

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If you have a query concerning the implementation or updating of this document, please contact the Owner on page 1